

Ashraf Syed

AI/ML Engineer | Generative AI | Data Science | MLOps Analyst

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SUMMARY

Collaborative AI/ML Engineer with 4+ years of experience contributing to machine learning, NLP, Generative AI, and MLOps initiatives across enterprise environments. Skilled in Python, SQL, PyTorch, Transformers, LangChain, Hugging Face, and modern cloud AI platforms. Experienced supporting cross-functional teams in building data pipelines, training ML models, fine-tuning LLMs (LoRA/PEFT), developing RAG systems, integrating vector databases, and implementing model monitoring, drift detection, and CI/CD workflows. Recognized for improving operational efficiency, automating workflows with GenAI, and delivering scalable ML solutions aligned with product, engineering, and business goals.

CORE SKILLS

- **Programming & Scripting:** Python, R, SQL, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Keras
- **LLMs & Generative AI:** Transformers, Hugging Face, LangChain, LlamaIndex, OpenAI GPT-4, LLaMA, Mistral, sentence-transformers Fine-tuning (LoRA/PEFT), prompt engineering, embeddings, vector search RAG pipelines, semantic search, retrieval optimization
- **Machine Learning:** Classification, regression, clustering, time-series forecasting, anomaly detection Deep Learning (CNNs, RNNs, LSTMs, Transformer architectures) A/B testing, model evaluation (precision/recall/F1/ROC-AUC), explainability (SHAP/LIME)
- **Vector Databases:** FAISS, Pinecone, ChromaDB, Weaviate
- **MLOps & Deployment :** MLflow, Docker, Kubernetes, Airflow, Prefect, GitHub Actions, CI/CD Model tracking, model registry, drift detection, pipeline automation FastAPI, Flask, REST APIs
- **Cloud Platforms :** AWS (S3, Lambda, Glue, RDS, SageMaker), Azure (Data Factory, Cognitive Services, Azure OpenAI), GCP (Vertex AI, BigQuery)
- **Data Engineering & BI:** ETL/ELT, SQL pipelines, dbt, PostgreSQL, MySQL, SQL Server Power BI, Tableau, Excel (PivotTables, XLOOKUP)

PROFESSIONAL EXPERIENCE

Intel | Product Analyst – AI/ML Focus

Stamford, CT | Jun 2025 – Present

- Collaborate with data scientists and ML engineers to refine anomaly detection and forecasting models using scikit-learn, XGBoost, and PyTorch, contributing to an 18% improvement in accuracy.
- Support development of Python and SQL pipelines orchestrated with Airflow to prepare ML-ready datasets, helping reduce feature engineering time by 35%.
- Co-develop LLM-powered internal tools using LangChain, OpenAI GPT-4, embeddings, and vector search (FAISS) to automate reporting and research workflows, lowering manual effort by 25%.
- Assist in building Power BI dashboards that monitor model performance, drift, and LLM usage metrics, improving issue resolution speed by 20%.
- Participate in A/B testing and experimentation to evaluate ML-driven recommendations and interface variants, contributing to a 15% increase in user conversions.
- Work with product managers and AI teams to define requirements for scalable Generative AI features including RAG pipelines, prompt workflows, and model evaluation protocols.

Alliance Technology Group | AI Business Analyst

Remote, US | Dec 2024 – Jun 2025

- Led design and deployment of LLM-based reporting automation using Azure OpenAI and LangChain, saving 30+ hours/month.
- Engineered ETL pipelines on AWS (S3, Lambda, Glue) for preprocessing, validation, and automated retraining.
- Built MLflow and CI/CD workflows for versioning, experiment tracking, and automated retraining.
- Developed KPI dashboards and drift detection frameworks in Power BI, improving data quality checks by 22%.
- Worked with data scientists to fine-tune GPT models and optimize prompts, improving contextual accuracy by 28%.

- Contributed to NLP pipelines using Python, spaCy, and Hugging Face Transformers (BERT/T5/BART) to extract insights from service logs and incident reports.
- Collaborated with ML engineers to develop classification models using XGBoost and BERT, reducing manual ticket triage time by 35%.
- Supported development of preprocessing and feature engineering workflows using Pandas, PySpark, and SQL, improving model training efficiency by 40%.
- Created dashboards in Tableau tracking model accuracy, SLA trends, drift indicators, and ticket clustering, enhancing visibility by 45%.
- Assisted in developing LLM-based summarization experiments using T5 and BART for automated RCA generation and ticket summarization.
- Helped implement SQL-based QA checks, anomaly detection rules, and data validation steps for reliable ML datasets.
- Worked with DevOps teams to containerize ML workflows using Docker and Kubernetes, improving reproducibility and environment consistency.

PROJECTS

LLM Powered Knowledge Assistant (RAG System)

- Developed a retrieval-augmented LLM chatbot using LangChain, OpenAI GPT, FAISS, and embeddings to search across 25,000+ enterprise documents.
- Designed chunking strategies, prompt templates, and caching layers to improve retrieval accuracy by 38%.
- Deployed via FastAPI with RBAC; evaluated outputs using ROUGE, BLEU, and human evaluation.

Healthcare Claims Forecasting & Anomaly Detection

- Built forecasting models (ARIMA, Prophet, XGBoost, LSTM) on 1.2M+ claim records.
- Automated ETL using AWS Glue and Airflow; developed Power BI dashboards for real-time insights.
- Enabled early detection of high-risk claim patterns and reduced reporting delays by 45%.

AI-Powered Ticket Summarization (NLP/LLM)

- Developed summarization pipelines using BART/T5, keyword extraction, entity recognition, and clustering.
- Preprocessed 500K+ records; integrated model via FastAPI to reduce ticket review time by 35%.
- Applied LoRA-based fine-tuning to improve summary relevance by 21%.

EDUCATION

Master of Science in Computer Science (focus on business analytics & product insights)
New England College, Henniker, NH, USA

Aug 2022 – May 2024